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1.INTRODUCTION

1.1 PROJECT OVERVIEW:

The Visualization Tool for Electric Vehicle Charge and Range Analysis project aims to analyze electric vehicle (EV) charging and driving range data using Tableau. The project focuses on understanding charging performance, battery capacity, vehicle range, charging station

availability, and other key factors that influence EV usage. Data preprocessing techniques such as handling missing values, removing duplicates, and formatting data were applied to ensure data quality. Interactive dashboards and visualizations were developed to identify trends, compare vehicle performance, and provide meaningful insights. The project helps users, manufacturers, and policymakers make informed decisions by presenting complex EV data in a clear and interactive visual format.

1.2 PURPOSE:

The purpose of this project is to develop an interactive visualization tool for analyzing electric vehicle charging and driving range data using Tableau. It aims to provide clear insights into battery performance, charging efficiency, vehicle range, and other key factors through interactive dashboards. The project helps users identify trends, compare vehicle performance, and support data-driven decision-making for the effective adoption and management of electric vehicles.

2. IDEATION PHASE:

2.1 Problem Statement

Electric vehicle users often face difficulties in understanding charging performance, battery capacity, vehicle range, and charging station availability due to scattered and complex data. This project addresses these challenges by creating an interactive Tableau dashboard that presents EV charging and range data in a simple, visual, and user-friendly manner for better decision-making.

2.2 Empathy Map Canvas

Says:

I want to know how far my EV can travel.

I need to find charging stations quickly.

I want reliable charging information.

Thinks:

Will my battery last for the journey?

Which EV offers better range?

How much time will charging take?

Does:

Checks charging station locations.

Compares vehicle range and battery capacity.

Uses dashboards to analyze EV performance.

Feels:

Concerned about battery range.

Wants convenient charging options.

Confident when accurate information is available.

2.3 Brainstorming

The project idea was developed by identifying the need for a visual platform that simplifies electric vehicle charging and range analysis. Tableau was selected to build interactive dashboards that help users compare EV performance, monitor charging efficiency, and identify useful insights through charts and filters.

3. REQUIREMENT ANALYSIS:

3.1 Customer Journey Map

Awareness: User wants information about electric vehicles.

Research: User explores charging and range data.

Analysis: User interacts with Tableau dashboards and filters.

Decision: User compares EV performance and selects a suitable option.

Outcome: Better understanding of electric vehicle charging and driving range.

3.2 Solution Requirement

Hardware: Laptop/PC, Internet connection.

Software: Tableau Public/Desktop, Microsoft Excel, Windows OS.

Dataset: Electric Vehicle Charge and Range dataset.

3.3 Data Flow Diagram

Dataset → Data Preprocessing → Tableau → Dashboard & Visualizations → User Analysis → Decision Making

3.4 Technology Stack

Tableau Public

Microsoft Excel

Windows Operating System

EV Charge and Range Dataset

4.PROJECT DESIGN:

4.1 Problem Solution Fit

The proposed Tableau dashboard effectively transforms raw EV data into interactive visualizations, enabling users to understand charging efficiency, vehicle range, and battery performance quickly and accurately.

4.2 Proposed Solution

Develop an interactive Tableau dashboard that visualizes electric vehicle charging and range data using charts, graphs, KPIs, and filters. The dashboard provides meaningful insights to help users compare EV performance and make informed decisions.

4.3 Solution Architecture

EV Dataset → Data Cleaning → Tableau Data Import → Dashboard Development → Interactive Visualizations → User Insights

5.PROJECT PLANNING & SCHEDULING:

5.1 Project Plan

Collect EV dataset.

Preprocess and clean the data.

Import data into Tableau.

Create charts and dashboards.

Add filters and calculated fields.

Test the dashboard.

Publish the dashboard and prepare the report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing was conducted to ensure that the Tableau dashboard performs efficiently with the EV dataset. The dashboard was tested for loading speed, filtering, and visualization accuracy.

Testing Results:

Dashboard loaded successfully without errors.

Charts and graphs displayed correct information.

Filters updated visualizations quickly.

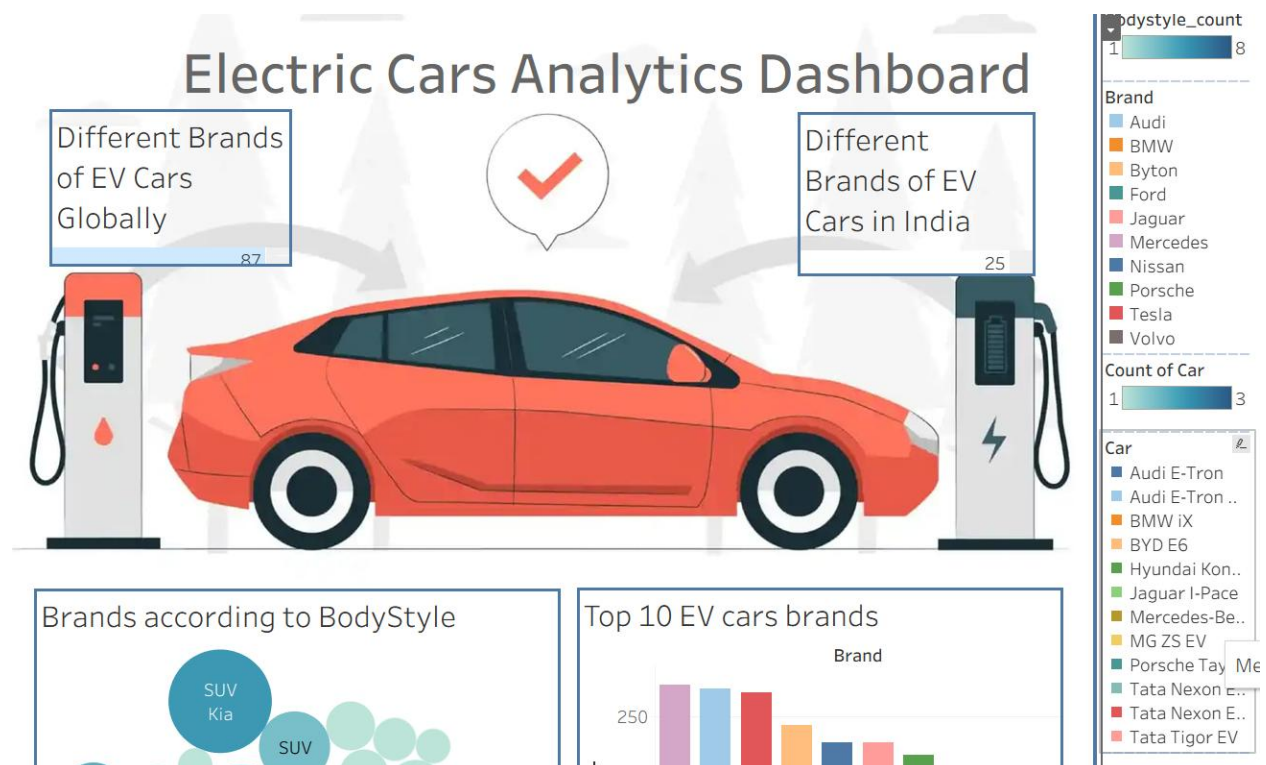
Data accuracy was verified after preprocessing.

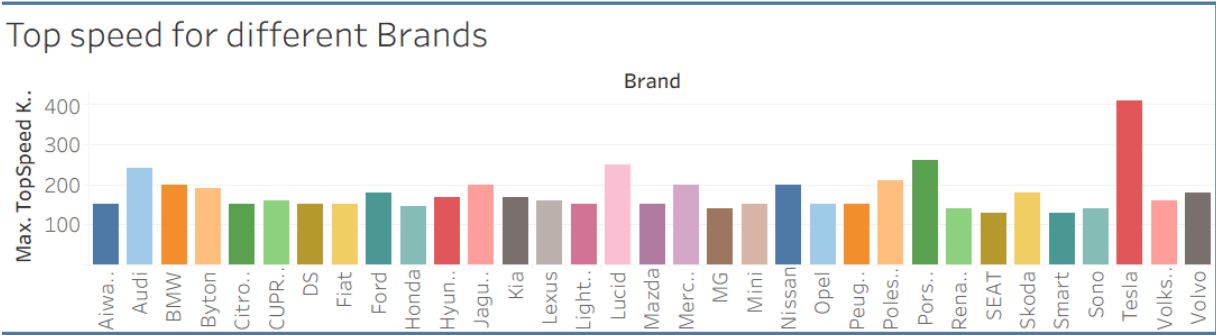
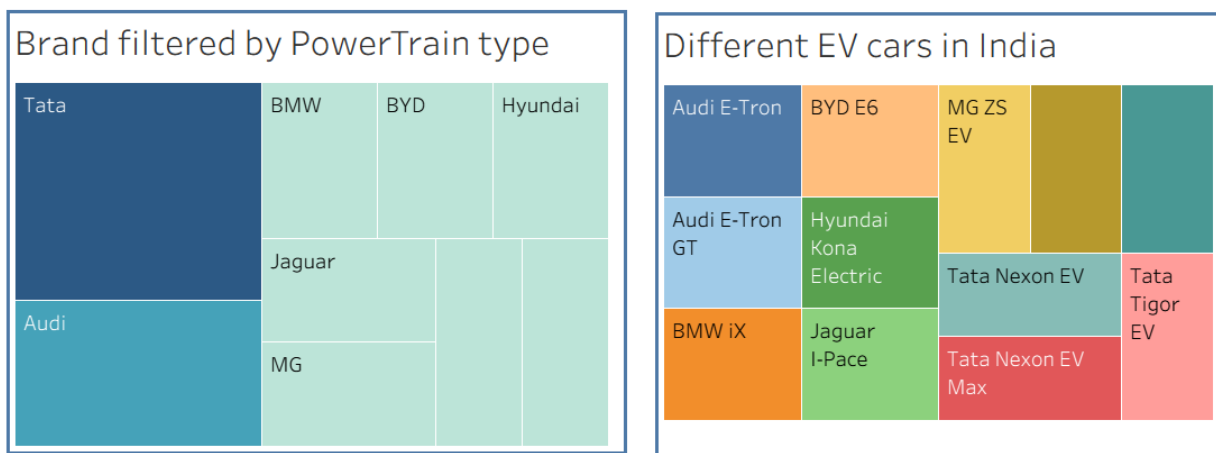
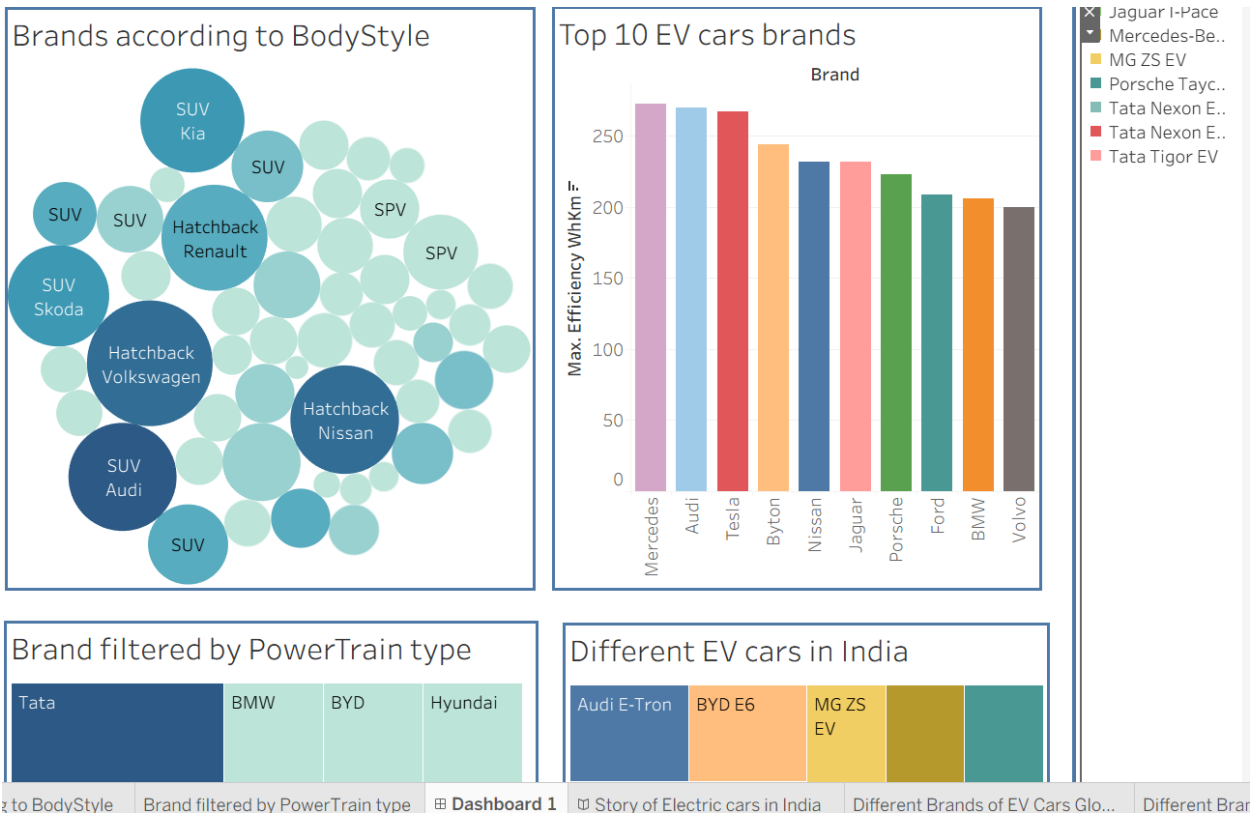
Dashboard performance remained stable while interacting with multiple charts.

7. RESULTS

The developed Tableau dashboard successfully visualizes Electric Vehicle charging and range analysis. Users can compare EV models, manufacturers, battery capacity, charging time, and driving range through interactive charts and filters.

7.1 OUTPUT SCREENSHOTS:





8. ADVANTAGES & DISADVANTAGES

Advantages

Easy to understand through interactive visualizations.

Helps compare EV charging time and driving range.

Supports data-driven decision making.

Fast and user-friendly dashboard.

Filters make analysis simple.

No coding is required to use the dashboard.

Disadvantages

Depends on the quality of the dataset.

Cannot provide real-time EV information.

Limited to the available data.

Large datasets may slightly increase dashboard loading time.

9. CONCLUSION

The "Visualization Tool for Electric Vehicle Charge and Range Analysis" provides an effective way to analyze EV data using Tableau. It enables users to understand charging performance, battery capacity, and driving range through interactive visualizations. The project simplifies complex data and supports better decision-making for users interested in electric vehicles.

10. FUTURE SCOPE

Add real-time EV charging station data.

Integrate live vehicle performance information.

Include predictive analytics for battery life.

Develop a web or mobile version of the dashboard.

Add more datasets from different countries.

Use AI/ML techniques to predict vehicle range and charging efficiency.

11. APPENDIX

Dataset Link:

<https://drive.google.com/drive/folders/1Rkzdk6Us1Uq2SRB4nxMAb83jN5bpHII?usp=sharing>

GitHub & Project Demo Link:

GitHub Repository:

<https://github.com/ShaikSaba07/Visualization-tool-for-electric-vehicle-charge-and-range-analysis/tree/main/1.%20Ideation%20Phase>

Project Demo Video:

<https://drive.google.com/file/d/1luYUs4API-CtF84l3NOD2IIJHD6Z4qxW/view?usp=drivesdk>